

João Caldeira

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Education

- Feb 2023 – Nov 2024 ♦ **M.Sc. COMPUTER SCIENCE AND ENGINEERING @ Instituto Superior Técnico, Universidade de Lisboa**
Thesis: ML-based trading system for stock market prediction; novel data labeling method with ensemble models specialized for short- and long-term trends.
- Sep 2018 – Feb 2023 ♦ **B.Sc. COMPUTER SCIENCE AND ENGINEERING @ Instituto Superior Técnico, Universidade de Lisboa**

Professional Experience

- Mar 2025 – Present ♦ **SOFTWARE ENGINEER & TECH LEAD @ Infosistema**
- ▶ Tech Lead and architect of *bizSupply*, designed a greenfield **microservices** B2B SaaS platform for enterprise document processing from scratch.
 - ▶ Engineered **agentic AI pipelines** combining document-understanding, **MCP servers**, and LLM agent tooling to automate end-to-end data extraction.
 - ▶ Built the infrastructure backbone: **event-driven pub-sub** messaging, **Redis** caching, and containerised cloud deployments with CI/CD automation.
- Jul 2023 – Mar 2025 ♦ **QA AUTOMATION ENGINEER @ SKY Portugal**
- ▶ Joined as a summer intern, built automated resilience testing frameworks, and secured a return offer to join full-time.
 - ▶ Delivered **API telemetry** and **CI/CD** pipeline improvements that reduced defect rates, streamlined testing workflows, and increased adoption and confidence across client teams.

Projects

- ♦ *Stock Market Prediction Research* – Peer-reviewed research on applying ML to algorithmic trading, yielding a profitable trading system outperforming key market indexes including the S&P 500 and NASDAQ-100. Published in Computational Economics (Springer, 2025).
- ♦ *SmartPenny* – Personal finance app that fully automates expense management, invoices are scanned, parsed via **OCR**, and categorised using an **LLM pipeline**.
- ♦ *Ricochet Robots Solver* – Solves a **Ricochet-Robots** game in the least amount of plays possible with optimal time-to-solution, using a **RBFS** in a search tree with a state data structure to store and represent the game information.

Skills

- Languages** ♦ Python · C# · Java · JavaScript · TypeScript · C++ · SQL
- Technologies** ♦ .NET · FastAPI · Spring · Next.js · React · PostgreSQL · Redis · Docker · Google Cloud · AWS · GitHub Actions · Vercel · Grafana · PyTorch · LLMs · RAG · MCPs · Agent Development Kit · Claude Code · Playwright · PyTest
- Soft Skills** ♦ Critical Thinking · Problem-Solving · Communication · Collaboration · Team Spirit